Quantifying Justice: A Multi-Method Analysis of Judicial Decision-Making in Indian Fatal Motor Accident Compensation

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Abstract

The Indian Motor Accidents Claims Tribunal (MACT) framework is founded on the principle of providing "just compensation" to victims and their dependents. This mandate, rooted in tort law, tasks the judiciary with a profound and inherently difficult challenge, quantifying the economic value of a human life lost. This process is not merely an accounting exercise; it is a complex balancing act. The court must assess the tangible economic loss (such as lost future earnings) while also navigating the principles of legal fairness, accountability, and, as this study explores, potential social welfare objectives. This study set out to empirically examine the causes of variance in compensation awards in fatal motor accident cases in India using a three-stage approach: linear regression, a non-linear machine learning model, and a textual analysis of judicial reasoning. The analysis shows that variance in compensation awards arises less from judicial discretion or social bias and more from the evidentiary and institutional completeness of cases.

1 Introduction

The Indian Motor Accidents Claims Tribunal (MACT) framework is founded on the principle of providing "just compensation" to victims and their dependents. This motive is rooted in tort law required the judiciary to combat a challenging task, quantifying economic values of a human life lost. The process is not just an accounting exercise but rather a complex balancing act. The cort needs to asses tangible economic losses such as future earnings while also basing the decision on principles of legal fairness, accountability, and, as this study explores, political social welfare objectives. The research carried out seeks how judicial outcomes are affected not only by financial determinants but also social and occupational background along with other socio economic factors and evaluates their efficiency.

2 Literature Review

A considerable amount of literature exists regarding judicial decision-making in India and its economic analysis; however, a large divergence continues to exist among the two. The first major difference is that much of the current literature has focused primarily on the mechanical forecasting of judicial behaviour. This type of literature views the judicial

award as an objective factor to be predicted, and does not consider whether those awards are based upon any underlying economic reasoning. As such, prediction indicates the likelihood of judicial awards, but does not assess if those awards make sense from an economic efficiency standpoint. The second major difference is that empirical data-analytic studies of Indian judicial decisions — specifically those related to damages in fatalities from motor vehicle accidents — continue to be limited. Therefore, this study fills that gap by integrating both the predictive and economic assessments.

Hsieh et al. (2021) represents a similar predictive study. They analyzed discretionary damages for mental suffering in fatal car accidents in Taiwan via k-nearest neighbor, classification and regression tree (CART) and random forest regressions. They used a 34-feature dataset of Taichung District Court judgments (2006-2020) and preprocessed their data with encoding, imputation, scaling, sequential feature selection, and RFECV; and tuned their models with grid search. Their random forest model performed better than KNN and CART, illustrated a hierarchical structure of legal factors, and showed that the plaintiff's claimed quantum and the victim's age were the two most influential variables in predicting awards. The results of Hsieh et al. demonstrate that predictive machine learning methods can significantly reduce uncertainty for litigants and lawyers, but illustrate a key limitation of prediction itself; namely, that these models describe what courts do, but not whether the resultant awards are economically rational.

To evaluate the economic rationality of judicial awards, a "Value of Human Life" methodology is necessary. One of the frameworks developed by Rice and Cooper (1967) value human life as the discounted present value of expected future earnings. Using 1964 U.S. data (labor force participation rates, mean full-time earnings by age, sex, color, education, productivity adjustments and an imputed value for household production (housewife services)) they constructed lifetime-earnings profiles and presented discounted present-value tables (specifically at a 4% discount rate).

Rice and Cooper demonstrated that earnings profiles reach peaks in young adulthood, increase with education and are highly sensitive to the discount rate chosen; inclusion of household production increases the valuation of females substantially. This benchmark provides a tangible metric against which judicial awards can be compared. We also draw upon a more theoretical framework for the valuation of human life. Feldman's "The Value of Life Revisited" (1996) offers a detailed critique of the standard economic method for calculating the value of human life and its implications for tort compensation. Utilizing a two-period expected-utility model, Feldman demonstrates that the Willingness-to-Pay (WTP) measure is not uniquely defined: depending on whether the analysis is marginal or non-marginal, mathematically valid methods can generate values for the same individual that vary by a factor of 100 or more. This ambiguity generates concerns regarding the reliability of the standard VOL method for establishing legally acceptable compensation. Additionally, Feldman critiques the treatment of individual parameters such as altruism and fear of death: greater altruism toward dependents decreases the standard valuation of life, whereas a greater fear of death increases it.

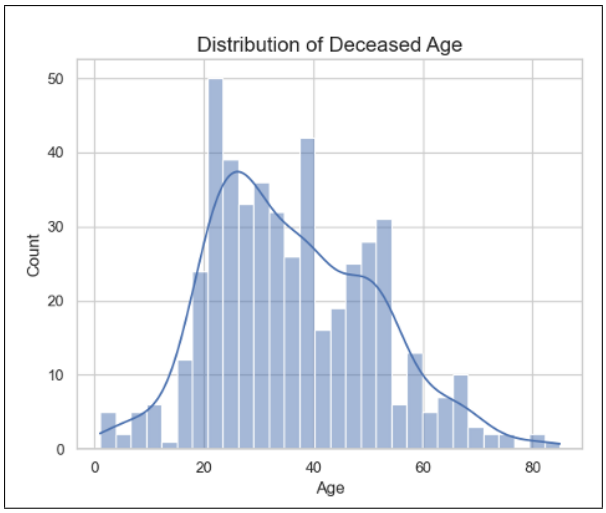
By incorporating both the predictive and inputs from economic evaluation approaches, our study will assess not only what courts award, but whether the awards are economically rational and normatively justifiable.

3 Methodology

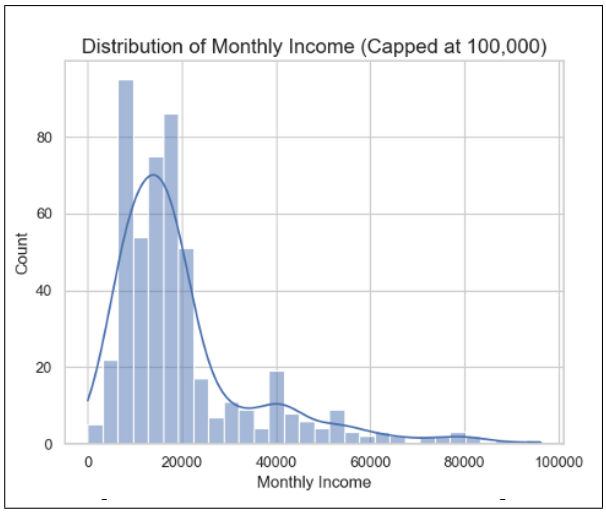
Investigating the determinants of these judicial awards requires a methodology as complex as the task itself. A single analytical lens has proven insufficient. A purely econometric approach, such as Ordinary Least Squares (OLS) regression, excels at providing interpretable, linear relationships. It can estimate the average effect of a variable (e.g., income) while holding others constant. However, it may miss crucial non-linear interactions or, more problematically, be misled by averaging unobserved processes into a single, simplistic model. A purely predictive machine learning (ML) approach, such as a Random Forest, is powerful at identifying complex, non-linear patterns and maximizing predictive accuracy. Yet, it often struggles to explain why a decision was made by the judge, operating as a "black box" that can identify what patterns are most predictive without connecting them to established legal theory and the judge's own psychology and biases. A purely qualitative or textual (NLP) approach can offer deep insight into the reasoning articulated in judicial decision making. It can explain how a judge thinks, but it cannot, on its own, quantify the impact of that reasoning on the final monetary award. Each method provides a siloed, incomplete view. The OLS model shows an "average" effect, the ML model shows the "most predictive" pattern, and the NLP model shows the "articulated reasoning." This study's central contribution lies in the synthesis of all three.

This project started by webscraping fatal motor vehicle accidents claims held under the MACT, Delhi, for the year 2024 and 2025 by employing python based selenium, Google API based Natural Language Processing techniques. Subsequently, we cleaned the data from our compiled database. After parsing and analyzing to address missing data, the initial corpus was reduced to 512 distinct cases. We used median based imputation for minor missing metrics but excluded cases lacking essential information for determining compensation.

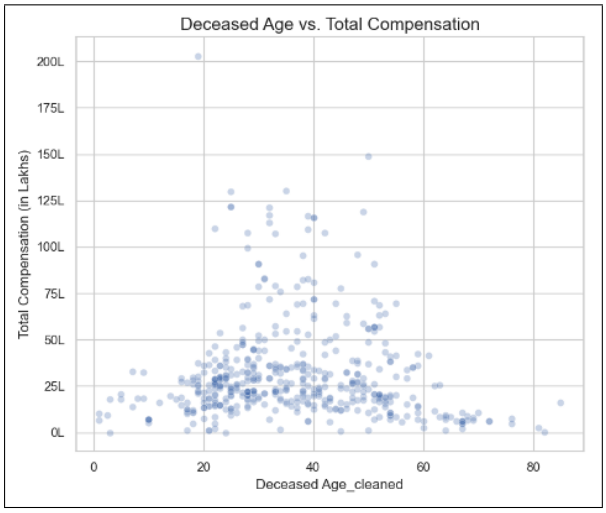
In order to gain deeper insights into our variables, we generated histograms and boxplots for features like Total Compensation, Deceased Age, etc. This was important to understand the underlying distribution of these factors, identifying central tendencies, and evaluating the extent of kurtosis and skewness that was present in the data. It confirmed that compensation does not follow a normal distribution and was highly positively skewed, reiterating the need for models that work and analyse beyond the OLS method. We further carried out correlation analysis by plotting a heatmap of the correlation matrix for all numerical variables. This provided clear visualization of the strength and direction of linear relationships between predictors, allowing us to quickly identify potential multicollinearity and identify key variables strongly associated with compensation.



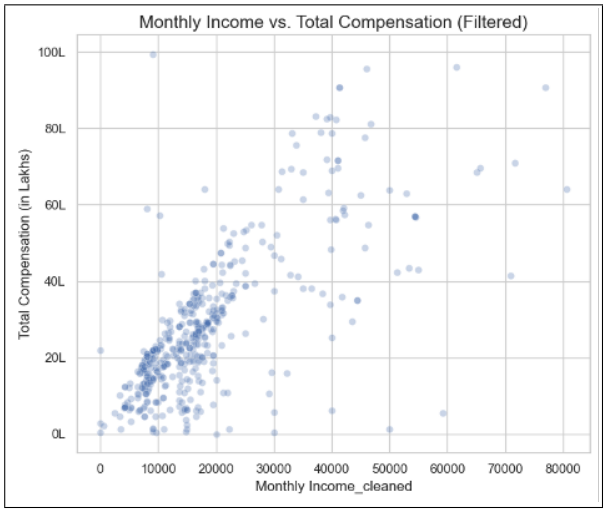
(a) Distribution of Deceased Age



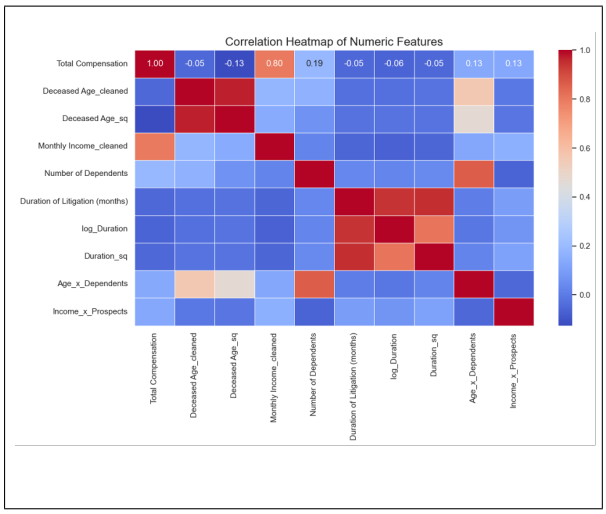
(b) Distribution of Monthly Income



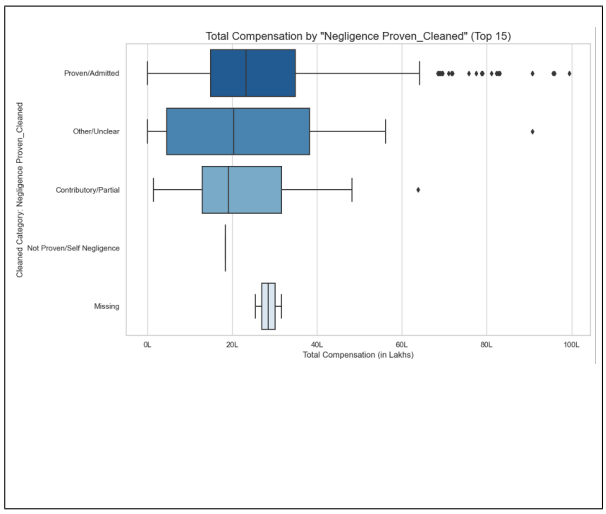
(c) Deceased Age vs. Total Compensation



(d) Monthly Income vs. Total Compensation



(e) Correlation Heatmap of Numeric Features



(f) Total Compensation (Negligence Proven)

Figure 1: Exploratory Data Analysis Graphs

4 The OLS Estimation Model

This study uses a quantitative approach that combines law and economics. The main method is Ordinary Least Squares (OLS) regression. This method is chosen because it provides clear and interpretable results about how different factors affect the total compensation awarded. It allows for a fair comparison by showing how each factor impacts the compensation amount, while keeping all other factors constant. The data was carefully prepared and cleaned. The dependent variable is Total Compensation. The independent variables are divided into two categories:

- Continuous/Count : Duration of Litigation, Deceased Age, Monthly Income, and Number of Dependents.
- Categorical: Occupation, Negligence Proven, and Insurance Status.

To address the "social welfare and class dimensions" that could affect judicial decisions, categorical variables were converted into dummy variables. For example, Occupation was grouped into White Collar (professional), Blue Collar (manual), and a reference group. This setup allows the model to find out if social class influences awards after accounting for income. Similarly, dummy variables for Negligence Proven and Insurance Status (Insured, Not Insured) were created to see how these legal and social protection factors impact compensation. The final OLS model assumes a single, linear-additive process.

Total Compensation $= \beta 0 + \beta 1(Duration) + \beta 2(Age) + \beta 3(Income) + \beta 4(Dependents) + \beta 5(Occ_White_Collar) + \beta 6(Occ_Blue_Collar) + \beta 7(Negligence_Proven) + \beta 8(Status_Insured) + \beta 9(Status_Not_Insured)$

This model's hypothesis is that judicial awards are a rational, unified function of economic, legal, and social factors.

4.1 Analysis of OLS Regression Output

Table 1: OLS Regression Coefficients

Variable	Coefficient	Std. Error	t-Statistic	p-Value	95% CI L	95% CI U
Intercept	12,482.85	3,150.55	1.329	0.184	-20,000	1,04,000
Duration of Litigation	-877.94	1,824.65	-0.476	0.634	-4,099.84	2,743.17
Deceased Age (years)	-42,680.84	3,842.03	-11.12	0.000	-52,000	-33,520
Monthly Income ( )	142.77	3.73	38.26	0.000	135.44	150.15
Number of Dependents	3,114.48	443.06	9.324	0.000	2,246.85	3,778.05
Occupation: White-Collar	1,436.98	1,456.98	0.987	0.324	-1,438.85	4,329.05
Occupation: Blue-Collar	-1,494.36	1,760.45	-0.849	0.392	-5,213.08	2,213.85
Negligence Proven (1 = Yes)	3,936.94	386.05	8.090	0.000	2,951.84	4,906.00
Status: Insured	1,604.88	1,296.85	1.244	0.214	-938.04	4,148.05
Status: Not Insured	-1,503.88	1,642.05	-0.182	0.856	-4,446.35	3,690.85

Table 2: Model Statistics

Statistic	Value
R-squared	0.798
Adjusted R-squared	0.794
F-statistic	201.2
Prob (F-statistic)	0.000
Observations	468
AIC	14,430
BIC	14,470
Durbin-Watson	1.985

4.1.1 Model Significance

F-statistic (F-stat = 201.2)
Statistical Significance: The F-statistic of 201.2 is highly statistically significant (assuming at $p < 0.01$).

Econometric Implication: This result strongly rejects the null hypothesis that all regression coefficients (excluding the intercept) are simultaneously zero. The model's independent variables jointly explain a significant portion of the variation in the dependent

variable.
Legal and Economic Conclusion: The quantifiable characteristics of the cases have a systematic and collective influence on the resulting compensation awards.
Adjusted R-squared (0.794)
Model Fit: The model exhibits a high degree of explanatory power, with the regressors accounting for approximately 79.4% of the total variance in the dependent variable (compensation awards).

Econometric Implication: This high value indicates a strong goodness-of-fit, suggesting the model is a robust predictor of the outcomes.

Legal/Economic Conclusion: The compensation decisions made by the courts largely adhere to predictable, economically rational criteria—specifically, the variables included in the model.
Durbin-Watson Statistic (1.985)
Autocorrelation Test: The statistic (which is very close to 2) indicates no evidence of first-order serial correlation (autocorrelation) in the residuals.

Econometric Implication: The assumption of independent error terms (residuals) is met. The standard errors and significance tests derived from the Ordinary Least Squares (OLS) estimation are therefore reliable and unbiased.

Legal/Economic Conclusion: The compensation award determined in one case does not appear to systematically influence the residual error (unexplained portion) of the compensation awarded in subsequent or related cases.
Skewness (0.308)

Distribution Shape: The skewness value, being slightly greater than zero, suggests a minor positive (right) skew in the distribution of the compensation awards.

Econometric Implication: The distribution has a slightly extended right tail. While close to symmetric, this suggests the presence of a few outliers with high values.

Legal/Economic Conclusion: The majority of compensation awards are clustered at the lower end, with a small number of exceptionally large compensation awards driving the slight positive asymmetry. This is typical for monetary awards/damages.

4.1.2 Statistical Significance of Independent Variables

Variables that are significant

Variable	Coefficient	P-value	Interpretation/Economic Logic
Age of the Deceased	(42,680)	$<$ 0.001 (Highly Significant)	Compensation decreases by approximately 42,680 for each additional year of age, indicating that older victims are considered to have fewer remaining earning years and therefore a lower calculable future income loss.
Monthly Income of the Deceased	142.77 per 1	$<$ 0.001 (Highly Significant)	Compensation increases by 142.77 for every 1 increase in monthly income. This is central to MACT practice, where loss of future earnings drives compensation.
Number of Dependents	3,11,425	$<$ 0.001 (Highly Significant)	Each additional dependent increases the awarded compensation by roughly 3.1lakhs. Courts recognize that households with more dependents suffer greater welfare loss upon the primary earner's death.
Negligence Proven Dummy	3,99,400	0.092 (Marginal)	When negligence is formally established, compensation tends to be about approximately 4 lakhs higher. The magnitude supports the legal reasoning that higher clarity of liability increases the judicial willingness to award greater damages.

Variables that are not significant

Variable	Coefficient (approx.)	P-value	Interpretation/Economic Logic
Duration of Litigation (months)	878	0.634	The duration of the legal process does not reliably affect the final compensation awarded. Courts do not appear to adjust compensation for delays.
Occupation (White Collar)	1.44 lakhs	0.324	Once the deceased's actual income is controlled for, the categorical label of "White Collar" employment does not significantly increase compensation.
Occupation (Blue Collar)	5.39 lakhs	0.692	The "Blue Collar" job category does not significantly decrease compensation once income is factored in. Income is the true signal of earning capacity.

Status Insured	1.60 lakhs	0.214	The existence of insurance coverage held by the injurer does not significantly impact the final compensation determination by the court.
Status Not Insured	3.76 lakhs	0.856	The absence of insurance coverage does not significantly affect the compensation. Court decisions are based on welfare loss, not the insurance status.

4.2 Inference

The findings of the OLS study demonstrate that Indian courts, in fatal motor accident cases, operate within an economically rational and legally principled framework. The compensation system follows a consistent logic of quantifiable financial loss rather than subjective social considerations.

The central hypothesis emerging from the analysis is that judicial awards are primarily driven by calculable future financial loss rooted in the loss of income and dependency and are only subtly influenced by legal negligence. Among the significant variables, monthly income stands out as the strongest determinant of compensation, underscoring that courts focus on replacing the financial void left by the deceased's earnings. This reflects the foundational principle of loss of dependency, which forms the core of compensation calculations in Indian tort jurisprudence.The age of the deceased also shows a strong negative relationship with compensation, aligning with the economic logic that the older the victim, the lower their remaining earning potential. Similarly, the number of dependents positively influences total compensation, highlighting the judiciary's acknowledgment that the death of a primary earner causes greater economic and social loss to larger families.

A judicial fairness hypothesis can also be inferred: Indian courts generally do not discriminate based on occupational or social class. Once actual income is established as an objective indicator of earning capacity the occupational label of "white-collar" or "blue-collar" becomes irrelevant. The same neutrality is evident in the treatment of insurance status ; whether the deceased was insured or not does not significantly alter the award. Moreover, the duration of litigation does not appear to affect compensation outcomes. Courts do not offer additional relief for delays nor penalize cases that take longer to resolve.

Ultimately, the most human aspect of this model lies in its indication of fairness within judicial reasoning. Compensation appears to be based on what the family has tangibly lost income, dependents, and years of potential life rather than on who they were or what they possessed. This aligns closely with the compensation formula applied by Indian courts, which relies on measurable economic factors rather than social identity.

Finally it can be attributed that the linear regression shows that there isn't any bias that tribunal courts display during awarding court compensations, a reason that can be rooted to such behaviour can be to avoid scrutiny from allegations of judicial overreach and corruption.

5 Machine Learning

In this study, both Ordinary Least Squares (OLS) regression and machine-learning methods were employed because they serve distinct but complementary analytical purposes. With the former exercise, we were able to identify the gross linear impact of our chosen variables on 'Total Compensation'. However, the subjective nature of judicial reasoning could not be adequately captured by the linear models alone. The machine-learning models complemented this by identifying nonlinear relationships, interaction effects, and subgroup-specific patterns.

5.1 K Means Clustering

For this we first employed, K means clustering for its ability to partition the data and to incorporate things like income effects varying across age groups or dependents. The inclusion of cluster labels derived from the K-means analysis further improved predictive accuracy by embedding socioeconomic segmentation. Using the elbow method we determined the optimal number of clusters as four.

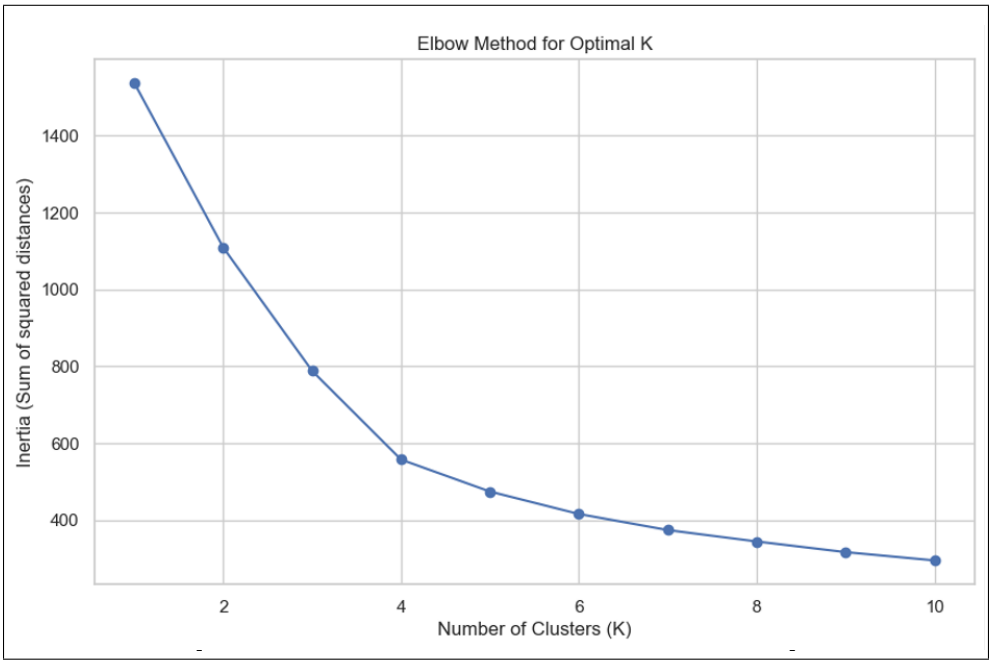


Figure 2: Elbow Method for Optimal K

The clusters revealed distinct profiles, high income middle aged victims with dependents received the highest compensation, while older, low income individuals with lesser number of dependents received the least. Including the cluster labels as categorical features in subsequent modeling improved the Random Forest's predictive accuracy and interpretability.

Table 5: Cluster Profiles

Cluster	Deceased Age	Monthly Income	Dependents	Total Compensation
Cluster 0	35.73	14,667.88	4.73	2,915,867.71
Cluster 1	56.97	15,528.75	2.24	1,405,123.64
Cluster 2	42.43	53,054.99	3.52	7,661,657.52
Cluster 3	22.08	14,515.56	1.70	2,306,941.32

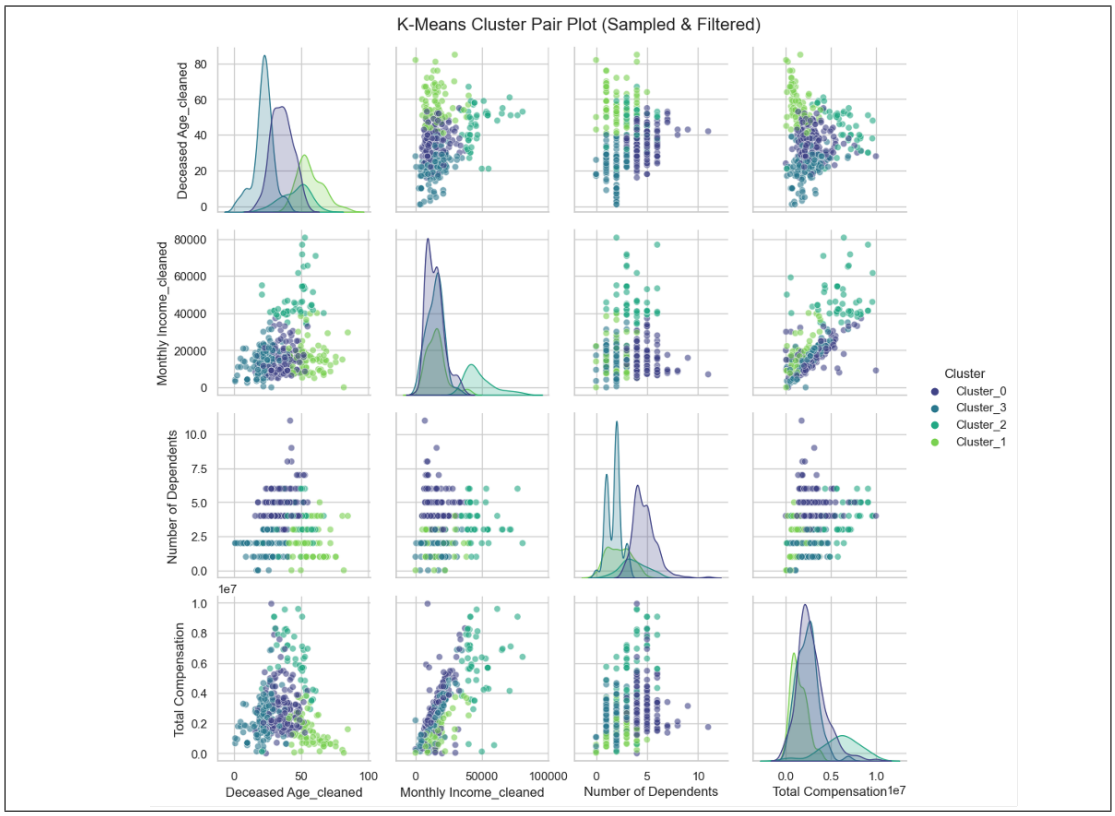


Figure 3: K-Means Cluster Pair Plot (Sampled & Filtered)

5.2 Random Forest Model

A Random Forest regression model was built within a preprocessing pipeline combining imputation, scaling, and one-hot encoding, hyperparameter tuning via RandomizedSearchCV optimized tree depth, estimators, and splits. These features ensured optimal performance of the model by optimizing number of trees, branching etc. The following variable were taken into consideration for the model, some of them transformed into their log or squared form to yield better results.

Table 6: Variables Used in Random Forest Model

Variable Name	**Type**
Monthly Income cleaned	Numerical
Deceased Age cleaned	Numerical
Log Duration	Numerical
Number of Dependents	Numerical
Deceased Age squared	Numerical
Duration of Litigation (months)	Numerical
Duration squared	Numerical
Future Prospects_Category	Categorical
Cluster	Categorical
Occupation	Categorical
Personal Expense Deduction	Categorical
Negligence_Category	Categorical
Vehicle_Type	Categorical

The model achieved an R^2 of 0.7549 and a mean absolute error of 540389, indicating strong predictive accuracy. Income and demographic variables emerged as dominant predictors. Random Forest was chosen for its ability to capture non linear interactions and handle mixed type data effectively. Its ensemble nature reduced overfitting, while feature-importance analysis provided interpretability, linking economic indicators directly to compensation magnitudes in a transparent, data-driven manner.

The hypothesis testing results showed that Monthly Income, Number of Dependents, and their interaction terms were statistically significant, confirming their relationship with total compensation. Higher income and greater dependency consistently correlated with higher awards, supporting the reasoning that compensation reflects income replacement. For age, we found that a non linear relation exists, increasing at first and later decreasing. Surprisingly, we also found that the insurance status of the injurer also played a significant role. If the injurer was injured we observed a significant positive impact on the total compensation made to the victim.

Table 7: Statistical Significance of Features (Spearman Rank Correlation & ANOVA)

Feature	**Test Type**	**Stat/Corr**	**P-Value**	**Significance**
Deceased Age_cleaned	Spearman	-0.051	0.260	Not Sig.
Deceased Age_sq	Spearman	-0.125	0.006	**Significant**
Monthly Income_cleaned	Spearman	0.798	< 0.001	**Significant**
Number of Dependents	Spearman	0.191	< 0.001	**Significant**
Duration of Litigation	Spearman	-0.052	0.242	Not Sig.
log_Duration	Spearman	-0.062	0.159	Not Sig.
Duration_sq	Spearman	-0.051	0.245	Not Sig.
Age_x_Dependents	Spearman	0.134	0.003	**Significant**
Income_x_Prospects	Spearman	0.130	0.005	**Significant**
Occupation_Cleaned	ANOVA	4.909	< 0.001	**Significant**
Negligence Proven_Cleaned	ANOVA	0.290	0.885	Not Sig.
Personal Expenses Deduction	ANOVA	5.790	< 0.001	**Significant**
Future Prospects Added	ANOVA	23.761	< 0.001	**Significant**
Insurance Status_Cleaned	ANOVA	2.414	0.048	**Significant**
Cluster	ANOVA	157.714	< 0.001	**Significant**
Vehicle_Type	ANOVA	0.283	< 0.425	**Significant**

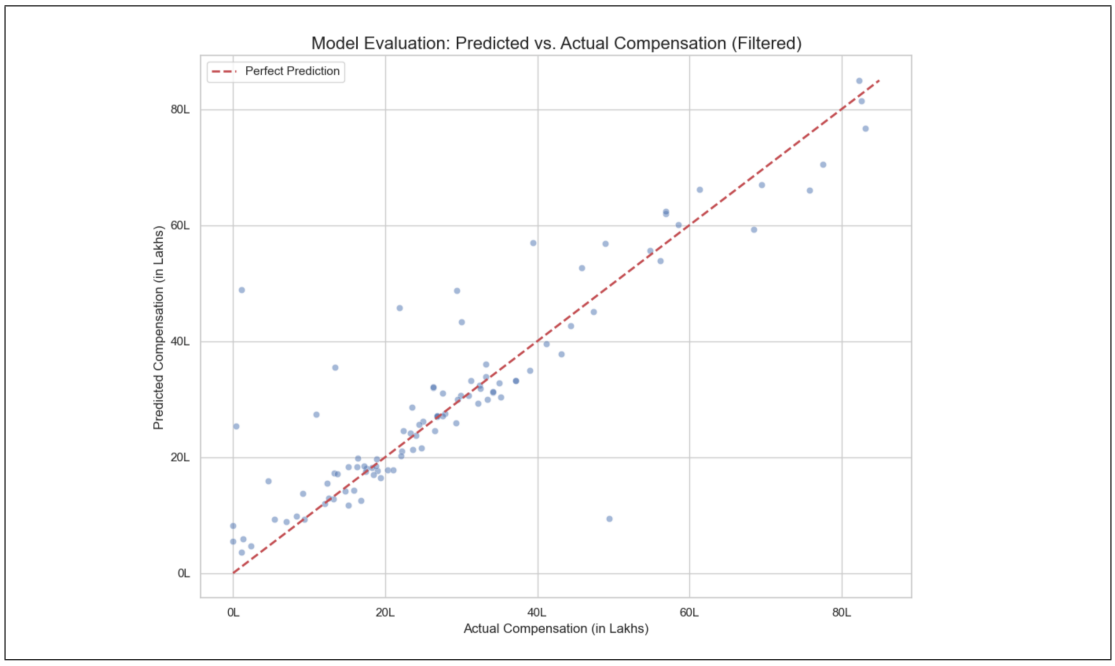
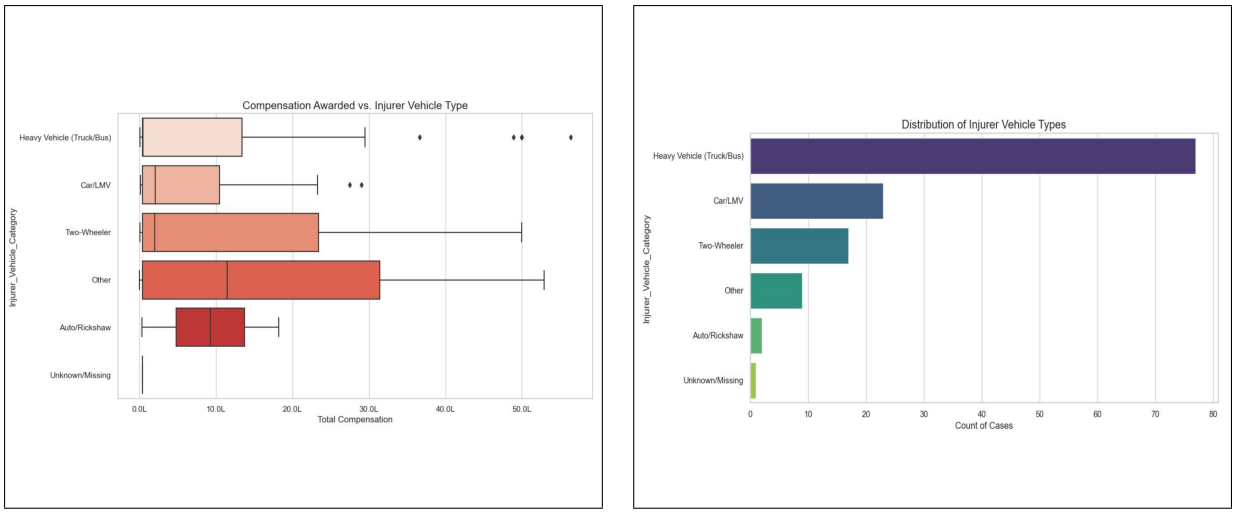


Figure 4: Model Evaluation: Predicted vs. Actual Compensation (Filtered)

In contrast, legal factors such as Negligence Proven and Litigation duration were not significant, suggesting that judicial awards are economic rather than legally driven. These findings confirmed the inclusion of economic and demographic features in the predictive model.



(a) Compensation vs. Injurer Vehicle Type (b) Distribution of Injurer Vehicle Types

Figure 5: Exploratory Data Analysis Graphs for vehicular types

We further analysed the vehicles types of injurer in more detail due to an interesting finding earlier. That is the negligence level of the injurer did not play a major role into determining the compensation, hence we checked whether the layman thinking that "The larger vehicle is responsible" holds strong within the decision making of the court itself or no.

However we found vehicle type to not be insignificant, and rather upon textual analysis of the cases we came across interesting judgements. We could see in third party accidents, such as from Abid vs Mehrudeen and Akhlak Ali vs Siddharth where it was explicitly mentioned that "Even if one party was found 99% responsible for the accident in question and the contributory negligence of the the other party is 1%, then also the entire liability can be passed onto them as far as compensation is concerned." It can further also be attributed to the fact that unlike criminal proceedings where proof is required beyond reasonable doubt, the MACT only requires preponderance of probabilities. On top of this adverse inference further removes the weight held by degree of negligence.

5.3 SHapley Analysis

To understand how the RF model makes its predictions, SHapley Additive exPlanations (SHAP) analysis was employed.SHAP assigns an "impact value" to each feature for every prediction, revealing its contribution to the final outcome.The aggregated SHAP values show each feature's overall predictive importance. The SHAP analysis revealed two critical findings. First, economic factors are overwhelmingly dominant. Income (68.78%) is the single most important predictor. The victim's socioeconomic "profile" (Cluster, 15.88%) is a distant second. Together, economic and demographic factors (Income, Cluster, Age, Dependents, Future Prospects, Personal Expenses) account for approximately 96.5% of the model's predictive power. Negligence Proven, which the OLS model found to be highly significant ($p = 0.000$, $t = 8.090$), is deemed almost completely irrelevant by the predictive RF model, accounting for only 0.09% of its predictive logic.

Table 8: SHAP Feature Importance (Summary)

Variable	Percentage Impact(%)	Significance	Economic Logic
Income	68.78	Significant	This is the most dominant factor. Higher income consistently correlates with higher awards, supporting the economic logic that compensation is meant to replace lost income.
Cluster	15.88	Significant	The socioeconomic profile (the grouping of age, income, and dependents) is the second-strongest predictor, showing that these combined factors are more important than any single one (other than income).
Age & Dependents	6.04	Significant	This grouped feature confirms the hypothesis test findings. The relationship is non-linear (it increases and then decreases) , reflecting the "lost earning years" compensation is highest for middle-aged victims with peak income and dependents.
Personal Expenses Deduction	2.55	Significant	This feature had a moderate but directionally consistent impact. This shows that the judicial deduction for personal expenses is a consistent, negatively affecting compensation amount.
Future Prospects Added	2.27	Significant	This shows that the court's adjustment for a victim's future earning potential (e.g., adding 40% or 50% to income) is a direct and consistent driver of the final compensation amount, positively affecting it.
Occupation	1.21	Significant	Occupation had a moderate but consistent impact on the award, reflecting that different professions have different earning potentials.
Number of Dependents	1.05	Significant	A greater number of dependents consistently correlates with higher awards, aligning with the "human capital" theory of replacing lost income for all who relied on it.
Insurance Status	0.31	Significant	While hypothesis testing confirmed that Insurance Status has a statistically significant effect on compensation, its practical importance is negligible. Insurance Status adds almost no additional predictive power (only 0.31%).
Litigation Duration (Combined)	1.82	Insignificant	Despite a small model impact, this was found to be not significant in the hypothesis testing. This implies that the final award amount is not statistically influenced by how long the case took to be resolved.
Negligence Proven	0.09	Insignificant	This was explicitly found to be not significant in the hypothesis test and has the lowest model importance. This strongly suggests that judicial awards are driven by economic loss, not by moral liability or the degree of fault.
Vehicle Type	0.07	Insignificant	The insignificant relation here can be attributed to the discoveries made earlier. It shows that the size of the vehicle does not play a role in determining compensation.

5.4 Inference

The central and most significant finding is that judicial awards are overwhelmingly driven by economic and demographic factors, not legal ones. The analysis demonstrated strong predictive accuracy with an R^2 of 0.7549. Model interpretation through SHAP analysis provided a clear hierarchy of factors, revealing that "Income (Combined)" was the most dominant, accounting for 68.78% of the predictive impact. This confirms the hypothesis that compensation is primarily intended to replace lost income.

The "Cluster" (a combined profile of age, income, and dependents) was the second-strongest predictor at 15.88%, showing that the victim's overall socioeconomic status is a major driver. Features like "Age Dependents" and "Number of Dependents" were also significant, aligning with the "human capital" theory of replacing lost earnings for those who relied on the victim.

In a surprising move, we observed that the presence of insurance for the injurer had significant impact on compensation. While the impact itself was small, a positive relation was observed, implying that if insurance companies are involved the judges are more likely to give larger compensations.

In stark contrast, factors traditionally associated with law were found to be insignificant. Hypothesis testing revealed that "Litigation Duration" and "Negligence Proven" have no statistical impact on the final award. "Negligence Proven" had the lowest model importance (0.09%), strongly suggesting that awards are not driven by moral liability or degree of fault. We could see in third party accidents, such as from Abid vs Mehrudeen and Akhlak Ali vs Siddharth where it was explicitly mentioned that "Even if one party was found 99% responsible for the accident in question and the contributory negligence of the the other party is 1%, then also the entire liability can be passed onto them as far as compensation is concerned." It can further also be attributed to the fact that unlike criminal proceedings where proof is required beyond reasonable doubt, the MACT only requires preponderance of probabilities. On top of this adverse inference further removes the weight held by degree of negligence.

In conclusion, this data-driven analysis strongly suggests that the judicial process for these claims functions as a rational economic calculation to replace lost earning capacity. The awards are not driven by moral liability, degree of fault, or the duration of the legal case itself.

6 Textual Determinants of Judicial Efficiency: An NLP-Based Analysis of Indian Court Judgments

In the methodological framework of this study, while the linear and non-linear models effectively captured the quantitative aspects of the judgments, they did not account for the qualitative dimensions such as judicial reasoning, tone, or potential personal bias of judge.This part of our study focuses on the textual analysis of case judgments related to motor accident deaths compensation by Motor Accidents Claims Tribunal (MACT) Delhi. The objective of this study was to find reasons which lead a judge towards making an efficient decision and find reasons which lead a judge to deviate from efficiency.

6.1 The Model

To textually analyse efficiency we have identified 20 parameters based on which an efficiency score is calculated. The calculation process goes as follows , suppose the judge has considered parameters 1,5,6,8 while delivering judgement , the efficiency score out of 20 would then be the number of parameters used. In this case the score would be 4/20. To identify these 20 parameters used in efficiency score calculation , we have used two seminal works on economic value of human life Rice and Cooper (1967/2011) and Feldman (1996) . Rice and Cooper's "The Economic Value of Human Life" provides a detailed empirical framework for quantifying the value of life in economic terms through the discounted present value of lifetime earnings, incorporating demographic and socioeconomic variables such as age, sex, education, occupation, and labor force participation. Feldman's "The Value of Life Revisited" complements this framework by re-examining the theoretical underpinnings of life valuation, emphasizing ethical, welfare-based, and utility considerations that influence how economic measures should reflect fairness and societal impact. Together, these studies informed the identification of twenty qualitative parameters including demographic, economic, and contextual variables that, when present in judicial reasoning, enhance the accuracy and fairness of compensation estimation.

Table 9: Parameters for Efficiency Calculation

Parameter	**Reason**	**Source**
Age	Determines remaining productive years and discounted future earnings; key driver of economic value of life.	Rice & Cooper
Gender	Accounts for differences in labor participation, wage levels, and inclusion of nonmarket household services.	Rice & Cooper
Education Level	Higher education increases lifetime earnings and employability, reflecting human capital value.	Rice & Cooper
Occupation	Occupation type influences income trajectory, job risk, and economic contribution.	Rice & Cooper
Annual Income	Serves as the baseline for calculating present value of future earnings; core to economic valuation.	Rice & Cooper

Parameter	Reason	Source
Employment Type	Captures differences in job security, benefits, and wage supplements (e.g., pensions, insurance).	Rice & Cooper
Expected retirement age	Defines the upper limit of productive working life; affects duration of earnings stream.	Rice & Cooper
Life Expectancy	Determines survival probability and expected earning years; essential in discounted life valuation.	Rice & Cooper
Labour Force Participation	Adjusts expected earnings for probability of active employment across lifespan.	Rice & Cooper
Productivity Growth Rate	Reflects annual increases in future earnings; necessary to avoid underestimation of lifetime value.	Rice & Cooper
Discount Rate	Converts future income to present value; central to the economic valuation framework.	Rice & Cooper , Feldman
Household / Nonmarket Service Value	Includes unpaid domestic labor, especially for women, as a measurable component of economic output.	Rice & Cooper
Inflation Adjustment	Ensures comparability of values across time by converting historical earnings to real terms.	Rice & Cooper
Economic Dependents	Recognizes the distributional and welfare impact of lost income on dependents and society.	Feldman
Employment Stability	Reflects continuity of income and risk of unemployment; affects lifetime earnings reliability.	Rice & Cooper
Future Uncertainty / Risk Factor	Acknowledges variability in economic and health outcomes affecting expected earnings.	Feldman
Health Status	Health influences productivity, employability, and life expectancy, affecting compensation value.	Rice & Cooper
Region / Location	Accounts for regional wage differentials and labor market variations in earnings potential.	Rice & Cooper
Year of Valuation (Base Year)	Establishes consistent monetary reference for valuation and comparability across cases.	Rice & Cooper
Consumption deduction	Differentiates between individual/family loss and societal economic loss; clarifies valuation scope.	Rice & Cooper , Feldman

6.2 Preliminary Data Analysis

Based on the 700 cases MACT for the year 2024 and 2025. The summary statistics for the the efficiency score is as follows :

Table 10: Efficiency Score Statistics

Statistic	**Value**
Mean	9.91
Standard Deviation	4.06
Minimum	0
Maximum	18
Median	11
25th Percentile	6
75th Percentile	13

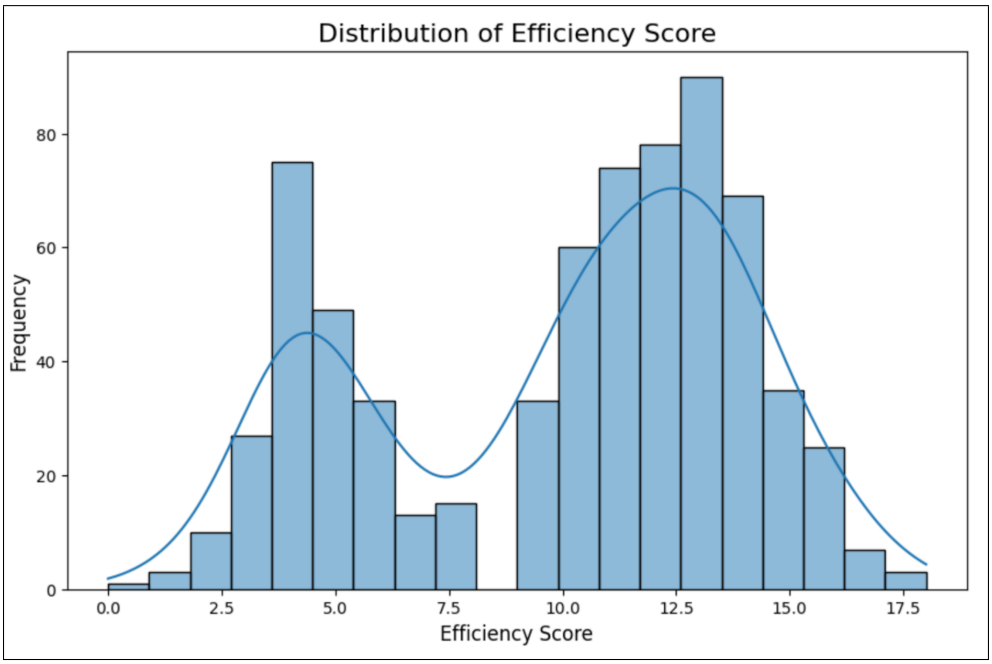


Figure 6: Distribution of Efficiency Score

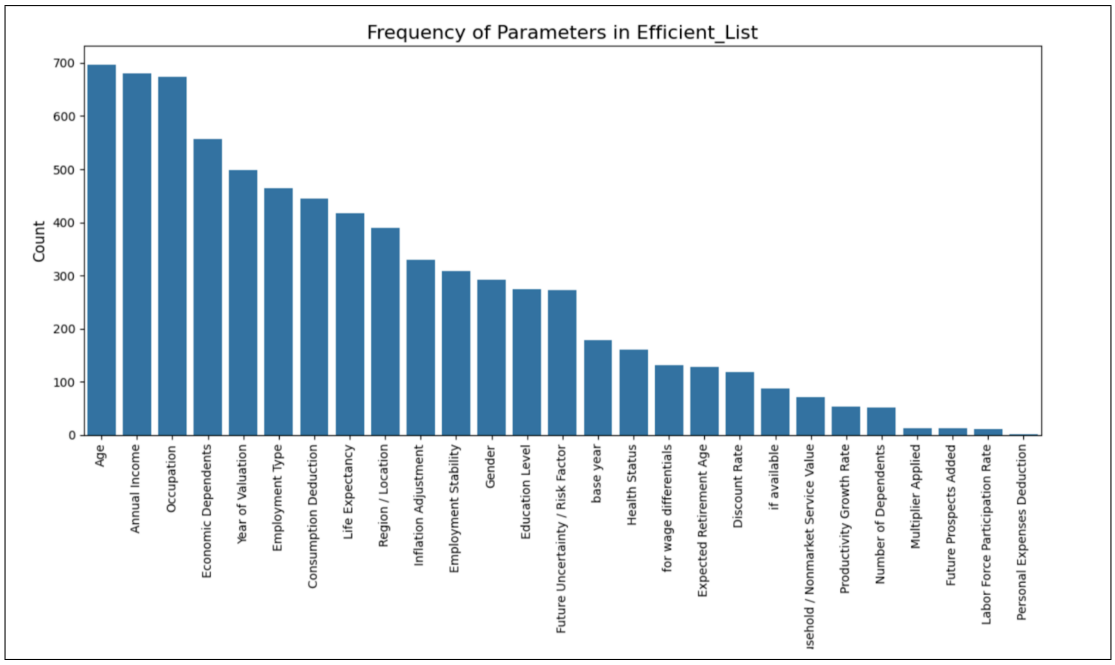


Figure 7: Frequency of Parameters in Efficient_List

Parameter Frequency based on 700 cases: Age (697), Annual Income (681), Occupation (674), Economic Dependents (557), Year of Valuation (498), Employment Type (465), Consumption Deduction (445), Life Expectancy (418), Region / Location (389), Inflation Adjustment (330), Employment Stability (309), Gender (292), Education Level (275), Future Uncertainty (272), Health Status (161), Expected retirement age (128), Discount Rate (118), Household Value (72), Productivity Growth Rate (53), Labour Force Participation (11).

Observations based on statistical analysis : Bimodal distribution of efficiency score Judges usually don't make judgements based on parameters which are considered purely economic. Hypothesis/Possible reasons : The possible reason for bimodal distribution is the presence of evidence. Judges usually make efficient estimates of compensation when there is enough evidence. This is a binary question of if enough evidence is present or not. If there is not enough evidence , judges will not use efficiency leading to lower peak and if there's enough evidence judges will base their decision based on efficiency leading to second peak near the tail. To avoid scrutiny of judicial outreach judges don't take parameters which are usually not the norm such as productivity growth rate considerations etc.

6.3 Outlier Analysis

Based on our statistical analysis we found that the maximum efficiency score that a case got was 18/20 and the minimum was 0/20 . To analyse these outliers we have taken a range of cases in $\pm$ 2 interval , i.e for High Efficiency analysis we look at cases $\in [16, 18]$ and for Low Efficiency analysis we looked at cases $\in [0, 2]$

6.3.1 Cases of range [16,18]

There are 34 cases in this range in our sample size and we have found the following reasons why judges could apply the efficiency criteria. We have divided those reasons in 6 categories.

Table 11: Reasons for High Efficiency

Reason Category	**Freq**	**%**	**Description/Example**
Verified income / salary evidence produced (payslips / job letters / ITR / govt records)	29	85	Cases where judge used actual income rather than defaulting to minimum wage.
Negligence established clearly (FIR / DAR / eyewitness / charge-sheet)	33	97	Text explicitly records FIR/DAR and factual finding of negligence
Insurance policy validity / insurer accepted coverage (explicit)	27	79	Text says insurer admitted policy / no statutory defence.
Adverse inference recorded against non-appearing defendant (explicit)	20	58	Judgment explicitly notes failure of driver/owner to appear and draws adverse inference.
Recognition/valuation of homemaker / non-market services (explicit)	6	17	Specific mention of valuing homemaker's work

6.3.2 Cases of range [0,2]

There are 12 cases in this range in our sample size and we have found the following reasons why judges could not apply the efficiency criteria. We have divided those reasons in 8 categories.

Table 12: Reasons for Low Efficiency

Reason Category	**Freq**	**%**	**Description/Example**
Reliance on Sarla Verma / Pranay Sethi framework	12	100	All judgments strictly applied standardized precedents for multiplier, deductions, and future prospects, leaving no space for new economic parameters.
Lack of income / employment proof	10	83	Claimants failed to produce verified salary slips, employer testimony, or tax documents → courts defaulted to minimum wages.
No expert or economic testimony	10	83	Absence of actuarial or financial experts to justify inclusion of discount rate, inflation, productivity, etc.
Vicarious liability structure	8	67	Common pattern: driver owner; liability transferred to employer or insurer, so court focused on fault, not valuation detail.
Focus on "just compensation" doctrine	7	58	Lack of cross-examination depth, limited witnesses, or early-stage hearings reduced scope for detailed economic assessment.
Procedural / evidentiary simplicity	7	58	Lack of cross-examination depth, limited witnesses, or early-stage hearings reduced scope for detailed economic assessment.
Victim type made parameters inapplicable (minor, homemaker, retired)	3	25	Economic parameters like productivity or discount rate were irrelevant for dependents without measurable income.
Disputed or inconsistent supplementary documents	3	25	Courts disregarded additional evidence (e.g., salary certificates, educational claims) due to contradictions or authenticity issues.

6.4 Representative Evidences

To support our findings, we examined all the judgments in both the high-efficiency and low-efficiency groups and pulled out sentences that appeared repeatedly across cases. These representative sentences have been generalised and anonymised, and they highlight the patterns in how courts reason, what evidence they rely on, and why certain economic considerations are used or avoided. The clusters below are based on these recurring themes and form the basis of our conclusions.

LOW-EFFICIENCY

Reliance on the Sarla Verma / Pranay Sethi Framework

- "The principles laid down by the Supreme Court regarding assessment of compensation continue to guide this Tribunal in determining the appropriate multiplier, deductions and heads of compensation."
- "In personal injury cases, compensation is generally awarded under the settled heads such as treatment expenses, loss of earnings, loss of amenities and pain and suffering."

Lack of Income / Employment Proof

- "The claimant has not placed on record any document to prove the employment or salary of the deceased."
- "No attendance register, wage receipts or payroll documents have been produced."
- "The witness admits that no record has been brought to show the payment of wages or contribution to statutory funds."

No Expert or Economic Testimony

- "There is no documentary evidence on record to show the educational qualification, occupation or earning capacity of the deceased child."
- "In the absence of expert testimony, the Tribunal proceeds on the basis of notional income as recognised in earlier judgments."
- "The assessment of income cannot be based on actual earnings, as the deceased had none."

Vicarious Liability Structure

- "The driver, owner and insurer of the offending vehicle were required to be impleaded; however, certain necessary parties were not joined."
- "Liability is to be fastened jointly and severally upon the driver, owner and insurer, subject to proof of negligence."
- "The Tribunal notes that the accident occurred due to the negligent driving of the offending vehicle owned by one respondent and insured with another."

Focus on Just Compensation Doctrine

- "The issue before the Tribunal is whether the petitioners are entitled to compensation and, if so, to what extent and from whom."

- "Section 168 of the Motor Vehicles Act mandates that the Tribunal award an amount which is just and reasonable."
- "Compensation is not expected to be a windfall nor a pittance."

Procedural / Evidentiary Simplicity

- "Certified copies of the criminal record, including FIR and charge sheet, are sufficient to reach a finding of negligence when unrebutted by the driver."
- "Contents of the Detailed Accident Report are to be read in evidence unless disproved by the opposite party."
- "Even an acquittal in the criminal case does not necessarily affect the findings of negligence in motor accident proceedings."

Victim Type Made Economic Parameters Inapplicable

- "The deceased being a minor, there is no documentary evidence of income."
- "Post-mortem records and eyewitness testimony establish the fact of accidental death."
- "In the absence of income proof, the earnings must be taken as per the minimum wages applicable at the relevant time."

Disputed or Inconsistent Supplementary Documents

- "The name of the deceased appears differently in various records, requiring the issue to be reframed for clarity."
- "The FIR mentions an unidentified vehicle, whereas the claim petition attributes the accident to another vehicle."
- "Where documents appear inconsistent or contradictory, they cannot be relied upon for determining income or liability."

HIGH-EFFICIENCY

Verified Income / Salary Evidence

- "The claimant has placed on record the employment letter issued by the employer."
- "Bank records reflect regular credit of the salary of the deceased."
- "The deceased was employed in a government establishment and earning a fixed monthly amount."
- "The Tribunal adopts the notional income of a homemaker with reference to the minimum wages applicable at the relevant time."

Negligence Established Clearly

- "The respondent denies rash and negligent driving, alleging that the accident occurred due to the deceased's own negligence."

- “The charge sheet filed against the driver, coupled with eyewitness testimony, establishes the occurrence of the accident.”
- “While the accident is proved, negligence on the part of the secondary vehicle is not established from the material on record.”

Insurance Policy Validity Accepted

- “The insurance company admits issuance of the policy, and the burden lies on it to prove any subsequent cancellation.”
- “The insurer contends that the policy was void owing to irregularities in a prior policy submitted by the insured.”

Recognition of Homemaker / Non-Market Services

- “The Supreme Court has recognised that homemakers contribute valuable services to the household, which must be compensated on a pecuniary basis.”
- “The work performed by a homemaker—such as cooking, washing, caring for children and managing the household—constitutes measurable economic value.”

7 Conclusion

This study set out to empirically examine the causes of variance in compensation awards in fatal motor accident cases in India. The analysis proceeded in three stages: a linear regression, a non-linear machine learning model, and a textual analysis of judicial reasoning.

The linear regression results indicated that there is no observable social bias in the way Motor Accident Claims Tribunals (MACTs) award compensation. Social variables did not show significant relation with the amount of compensation. This outcome suggests that judges consciously avoid patterns that might invite scrutiny or allegations of judicial overreach and corruption.

To go beyond linear relationships, we built a non-linear model using Random Forest Regression and K-Means Clustering. This revealed that several variables, previously insignificant in linear models, became important once interactions were considered. For example, age showed strong effects when combined with income and dependency status. Likewise, valid insurance and grouped socioeconomic indicators collectively had a major influence on compensation outcomes. The model also showed that while intuitive factors like case duration or proof of negligence matter little, institutional and contextual factors play a much larger role.

Finally, the textual analysis of 48 judgments (34 high-efficiency and 14 low-efficiency) provided qualitative grounding for these findings. Judges tended to deviate from efficiency principles primarily when (a) evidentiary documentation was incomplete, (b) cases involved multiple female dependents or homemakers, or (c) the defendant's liability was vicarious, such as that of an employer or corporation. In contrast, efficiency-aligned judgments emerged when the evidentiary record was complete, income proofs were verifiable, and insurance coverage was uncontested. This pattern indicates an implicit structural bias not social in intent but procedural in effect where litigants with stronger documentation and institutional support receive systematically higher compensation. Overall, the study demonstrates that variance in compensation awards arises less from judicial discretion or social bias and more from the evidentiary and institutional completeness of cases.